Finlay Phillips

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British Citizen — full right to work in the UK

PROFESSIONAL PROFILE

MEng Electronic and Electrical Engineering student at UCL looking for quant software and data-infrastructure roles. Interested in low-latency engines, market-data pipelines, execution systems, and production backends that keep state honest under failure.

EDUCATION

University College London 2024 â€“ 2028

MEng, Electronic and Electrical Engineering

- Coursework highlights: Stochastic Calculus and Uncertainty Analysis (Year 3), Engineering Mathematics in Finance (76%), Mathematical Modelling and Analysis I and II (75% / 76%), Advanced Engineering Mathematics (Year 3), Data Mining and Analysis (Year 3).
- 3rd-Year MEng Project: Hardware-Accelerated Limit Order Book (UCL, 2026â€“2027).

Strodeâ€™s College 2022 â€“ 2024

- A-Levels: Mathematics (A*), Physics (A), Further Mathematics (A). GCSEs: eight grades 9 to 7, including English Language (9) and Mathematics (8).

TECHNICAL SKILLS

- **Languages:** C++23, Python, Rust, TypeScript, SQL, MATLAB.
- **Infrastructure and Data:** PostgreSQL, Parquet, Polars, Docker, Linux/VPS, systemd, Azure, Git.
- **Developer Tools:** CMake, Google Test, Google Benchmark, Next.js.

TECHNICAL EXPERIENCE

Co-Founder and Lead Engineer — Toolbox Maths (EdTech startup backed by University of Southampton funding) 2024 â€“ Present

- Co-founded an EdTech startup building an interactive GCSE and A-Level maths platform with live student traffic and paid subscriptions.
- Built serverless Azure Functions APIs over Cosmos DB for learning aggregates, question banks, and conversation state, focusing on low-latency reads and consistent document updates.
- Implemented Stripe billing with server-side entitlement checks, signed webhooks, and idempotent billing-event queues so payment state stays correct under retries.

3rd-Year MEng Project: Hardware-Accelerated Limit Order Book — UCL 2026 â€“ 2027

- Designing and implementing a low-latency SystemVerilog engine on an FPGA to parse NASDAQ ITCH protocol market data streams.
- Reconstructing and maintaining deterministic limit order book state directly in hardware at wire speed.

tickforge, C++ Limit Order Book and Market-Making Engine — Personal project 2025

- Built an interactive, low-latency C++ trading simulator where users can watch buy and sell orders match in real time and test market-making strategies.
- Implemented a price-time priority limit order book with IOC and FOK order types, plus an Avellaneda-Stoikov market maker in a deterministic simulator.
- Kept the matching hot path allocation-light with an order arena and reserved standard-library containers, and measured latency with Google Benchmark.

Polymarket Market Data and Paper-Trading System — Personal project 2025

- Built a live data collection and paper-trading engine for prediction markets to test trading ideas without risking real capital.
- Ingested live order-book depth into zstd-compressed Parquet shards and ran serialized shadow jobs under a 750 MB Linux VPS memory limit.

Market Data Pipeline and Backtesting (Macro Bias) — Personal project 2025

- Built a daily cross-asset feature pipeline that lands market inputs in PostgreSQL and scores forecasts after the close.
- Used upsert transactions and state checks so failed ingestion jobs can be retried without duplicating rows.

ACTIVITIES

- **UCL Fintech Society:** Workshops and talks on algorithmic trading, fintech, and financial markets.